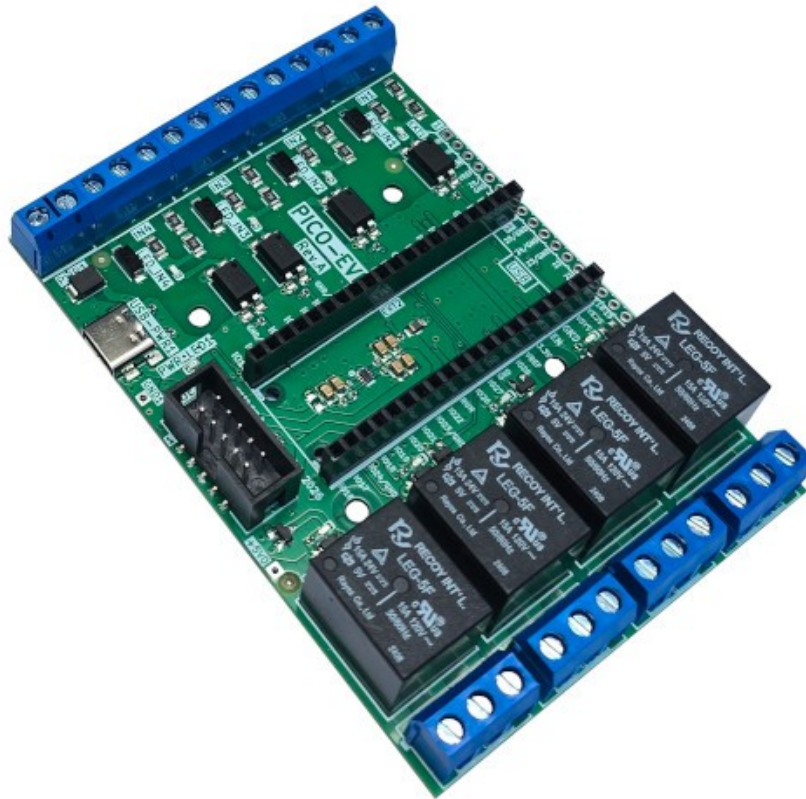




PICO-EVB

User Manual

olimex.com

Rev.1.0 July 2026

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Introduction to PICO-EVB

[PICO-EVB](#) is evaluation board which can be used with Raspberry Pi PICO, PICO-W, PICO2 and PICO2-W. It also can work with Olimex RP2040-PICO30.

[PICO-EVB](#) turns PICO into PLC and home automation device as it can drive high voltage loads with relays, and can sense via optoisolated inputs 3.3-30VDC or directly 110-220VAC signals.

[PICO-EVB](#) have this features:

- 40 pin DIL connector for PICO modules
- USB-C power supply connector which allow more current to be used by the board
- Four 15A/240VAC relays controlled by the PICO
- UEXT connector
- Four opto-isolated inputs which can work either with 3.3-30VDC or 110-220VAC
- Power supply can be feed to screw connector terminal for more robust connection
- All components are industrial grade -45+85C
- Dimensions 100x70 mm



Important notice: The Relays and high voltage inputs should be used with care.

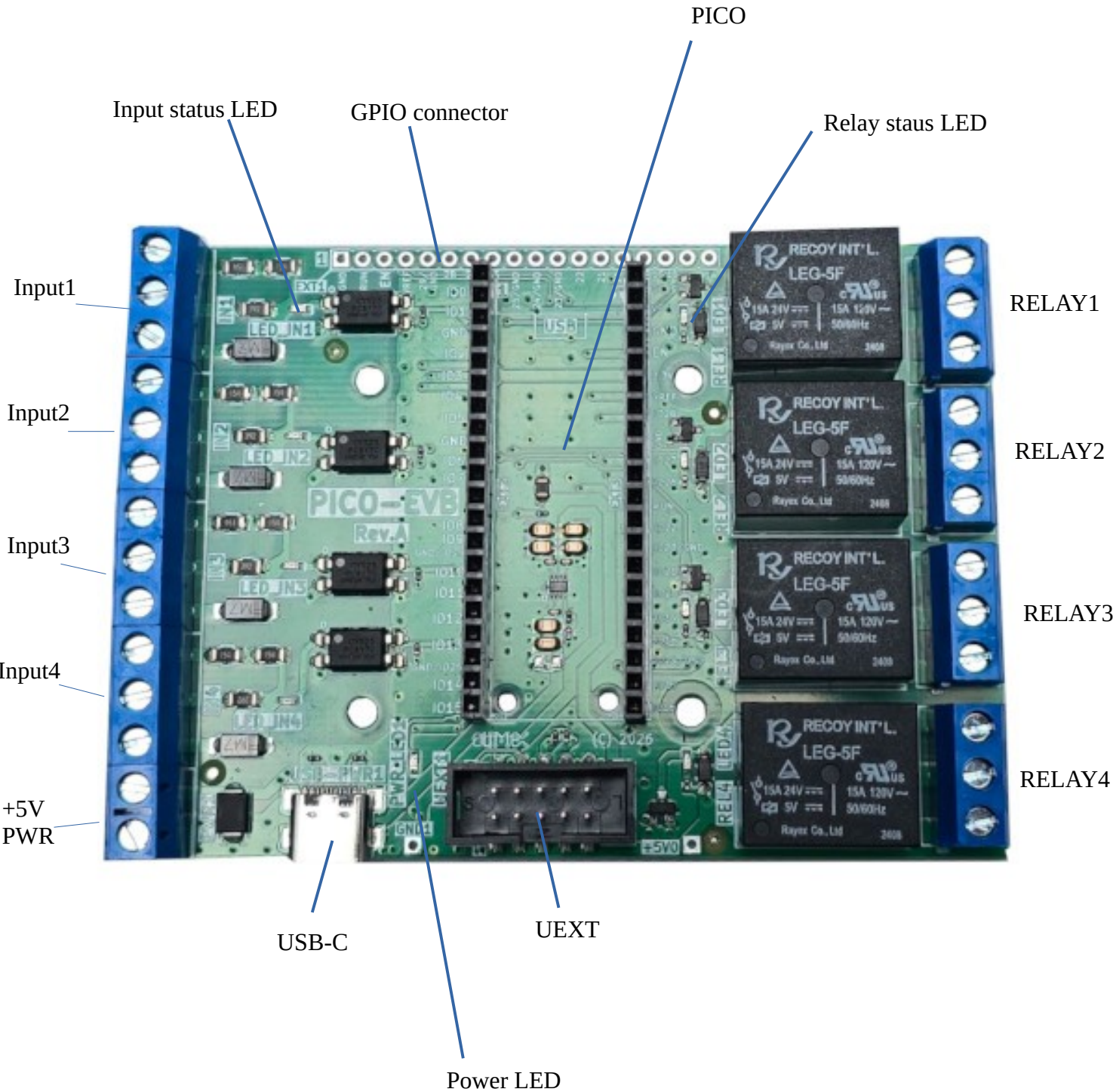
Each input have three connections: one common, one for +3.3-30VDC signals and one for 110/220VAC signals. If you want to drive the inputs with microcontroller feed 0 and +3.3V. If you want to drive the input with high voltage use live and neutral to connect. DO NOT CONNECT low voltage and high voltage to SAME input at the same time!

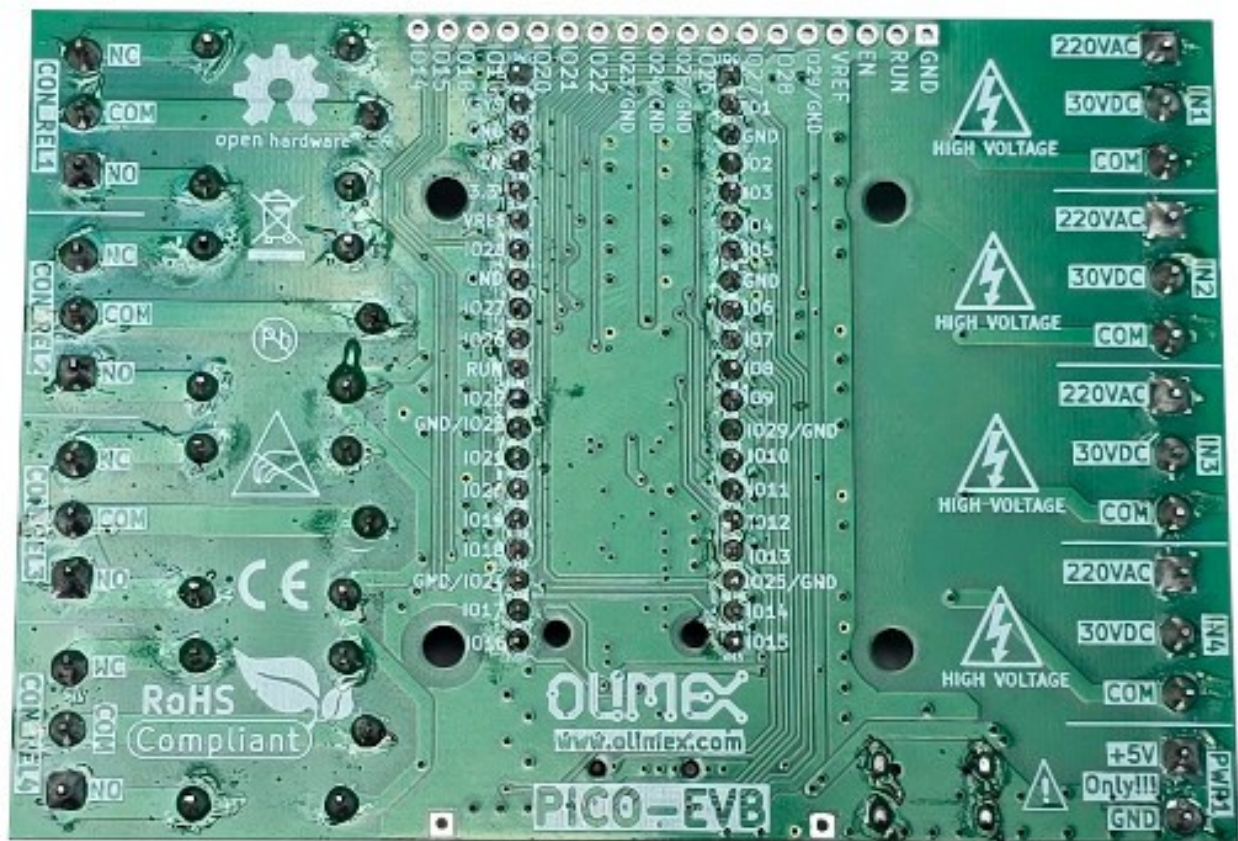
Order codes for PICO-EVB and accessories:

<u>PICO-EVB</u>	industrial grade PICO evaluation board
<u>USB-CABLE-A-TO-C-1M</u>	USB-C quality cable for data and power
<u>RP2040-PICO-HDR</u>	PICO with soldered headers
<u>RP2040-PICO-W-HDR</u>	PICO-W with soldered headers
<u>RP2350-PICO2-HDR</u>	PICO2 with soldered headers
<u>RP2350-PICO2-W-HDR</u>	PICO2-W with soldered headers

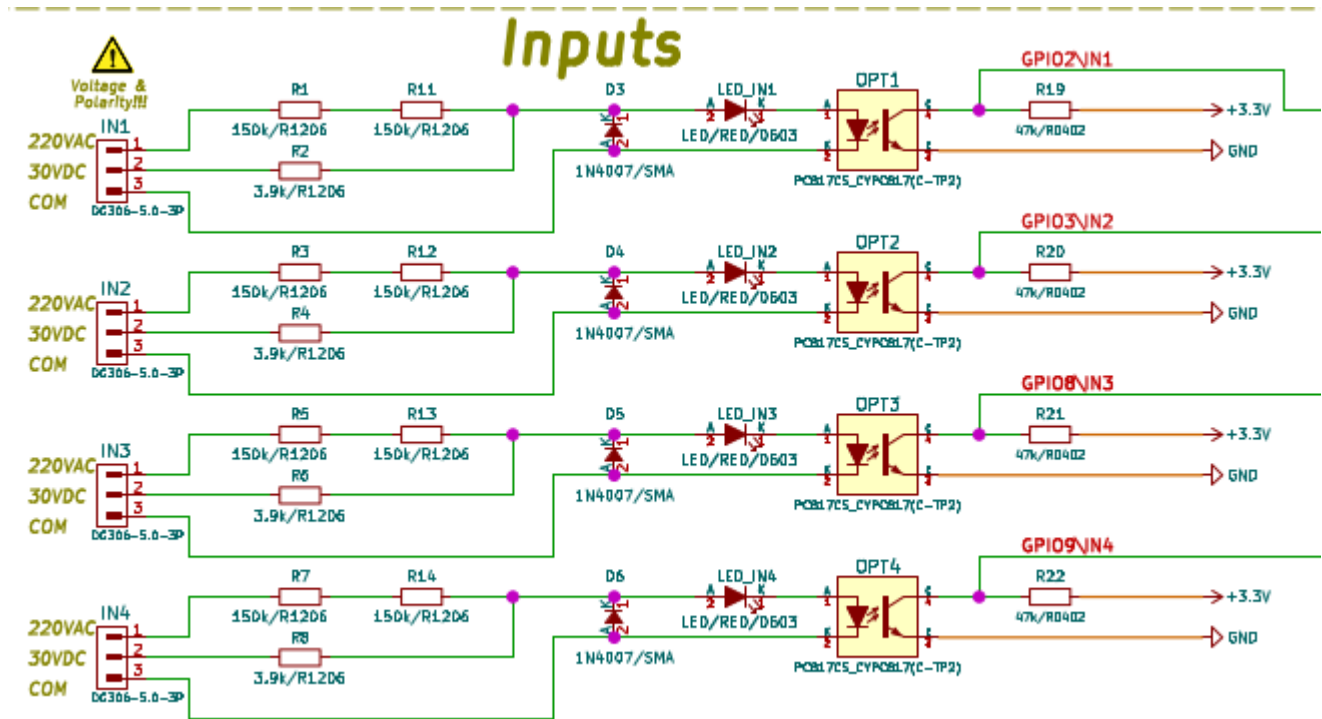
HARDWARE

PICO-EVB layout:

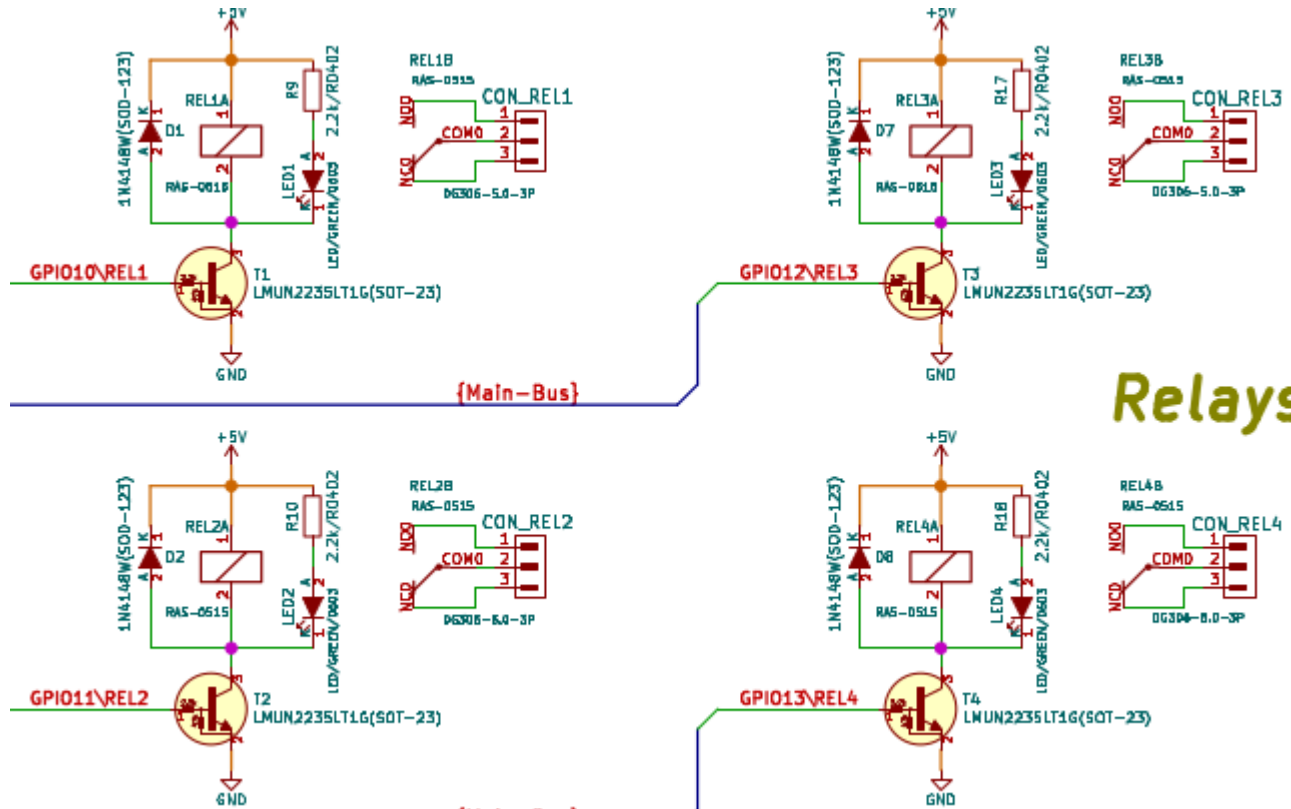




PICO-EVB inputs:



PICO-EVB Relays:



[PICO-EVB](#) can be powered by: USB-C connector, Terminal Block connector, PICO USB connector

You can connect any of them or all at the same time and it will be safe.



PICO-EVB schematics:

[PICO-EVB](#) latest schematic is on [GitHub](#)

UEXT connector:

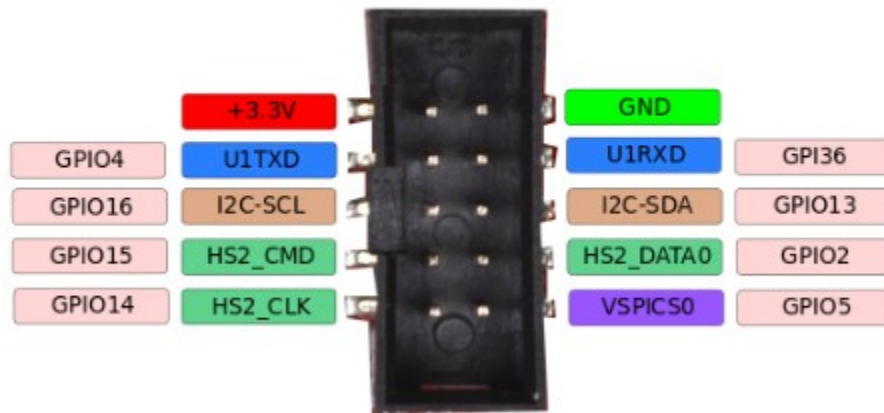
UEXT connector stands for Universal EXTension connector and contain +3.3V, GND, I2C, SPI, UART signals.

UEXT connector can be in different shapes.

The original UEXT connector is 0.1" 2.54mm step boxed plastic connector. All signals are with 3.3V levels.

UEXT connector

note it share same pins with EXT1 and EXT2



As the boards become smaller and smaller some smaller packages were introduced too beside the original UEXT connector

- mUEXT is 1.27 mm step boxed header connector which is with same layout as UEXT
- pUEXT is 1.0 mm single row connector (this is the connector used in RP2040-PICO30)

Olimex has developed number of [MODULES](#) with this connector. There are temperature, humidity, pressure, magnetic field, light sensors. Modules with LCDs, LED matrix, Relays, Bluetooth, Zigbee, WiFi, GSM, GPS, RFID, RTC, EKG, sensors and etc.

SOFTWARE:

[PICO-EVB](#) repository have these software:

- [Arduino-demo](#)
- [ESPHome](#)
- [MicroPython-demo](#)
- [MicroPython-server](#)

Revision History

Revision 1.0 July 2026